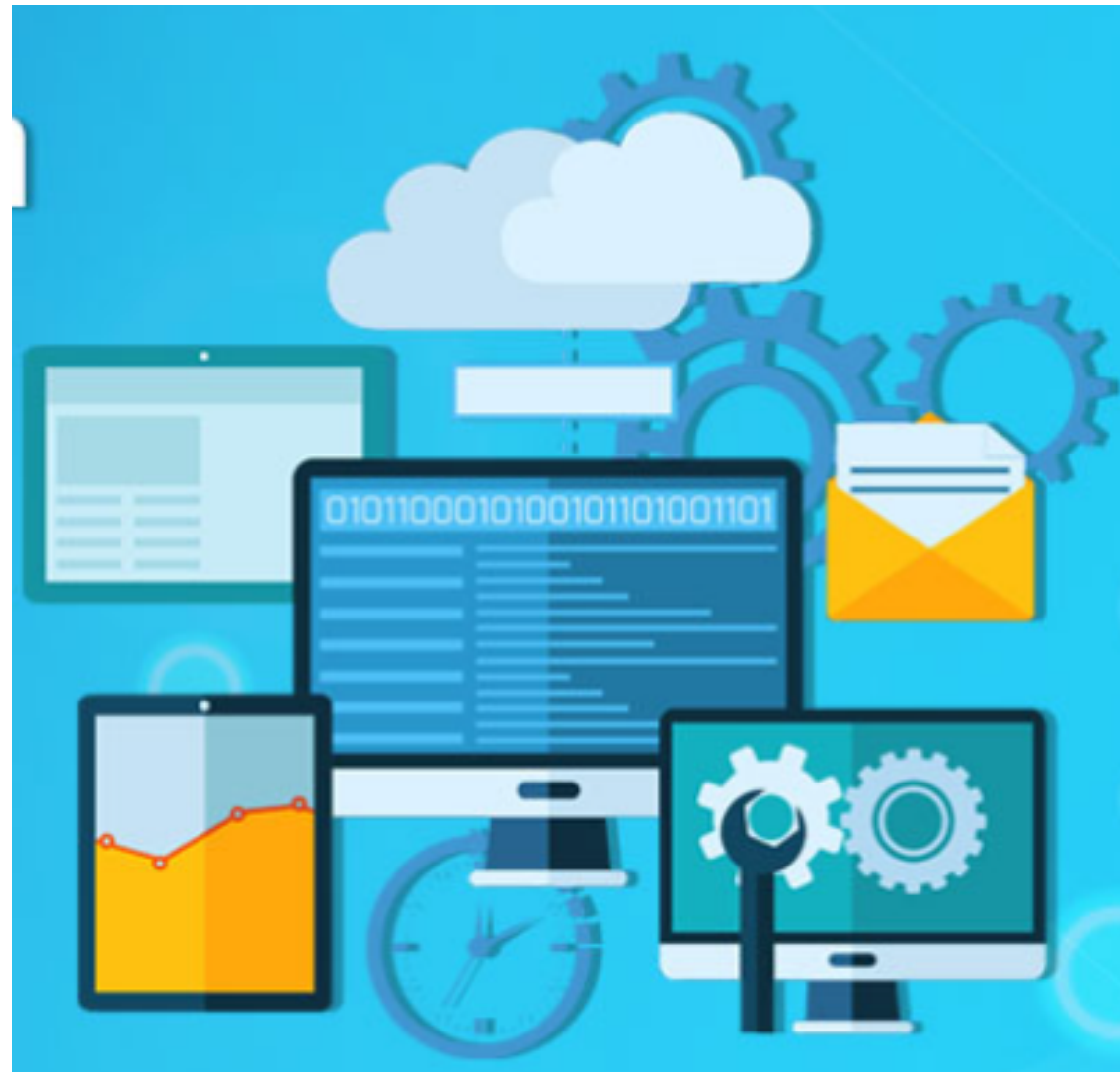


REQUIREMENT ANALYSIS

SALESTORM - SYSTEM DESIGN HACKATHON



DEPARTMENT : CSE-(AIML) - B

TEAM NAME : VISION WEAVERS

TEAM MEMBERS :
SANGHAMITHRA P
RUPASHREE V M
SHUBHAHARINI S

The Flash-Sale Dilemma: High-demand sales trigger massive, instantaneous spikes in traffic where thousands of buyers attempt to purchase a very limited supply simultaneously (e.g., 10,000 concurrent users competing for only 100 units).

Business Risk: Without proper design, this surge causes system crashes, database contention, duplicate charges, and—worst of all—overselling (selling more units than are actually available, leading to costly cancellations and broken customer trust).

The Core Business Problem

Core Business Question: How to process concurrent purchase spikes without crashing or overselling, while ensuring payment and order lifecycles remain reliable and recoverable when downstream services fail.

